Learning to Learn

Part 1: How Humans Learn

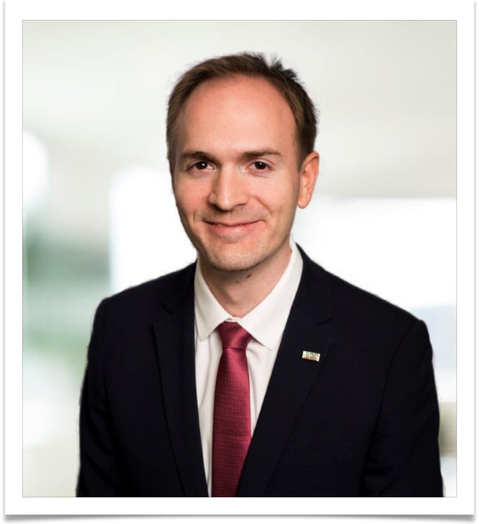
Introduce speakers



S.D. Nelson

Nelson is a PMP who contributed to testing robots at NASA. He was dubbed “Global Geek” for leading projects world-wide. His goal is to work in 100 countries because learning how other cultures solve problems adds more project management tools to his project management toolkit. Currently, he is putting his experience to work focusing on successfully adding emerging technologies to the project team. As project managers, what role can we assign to artificial intelligence? Can IoT Devices solve problems and report status like human team members?

He has an open source lab near Los Angeles that is conducting experiments alongside the I3 Consortium and the IOTA Foundation.



Bill Neil

Bill Neil is passionate about continuous learning and growth. He is a Certified Public Accountant turned Project Management Professional. As a Director at a large national public accounting firm, Bill leads development and change management efforts for implementing data analytics throughout the firm’s audit practice. This includes projects to design analytics dashboards used across thousands of client engagements as well as numerous professional development webinars, self-study courses, videos, informal trainings and feedback sessions for all levels of employees.

At PMI-OC, Bill volunteers as the chapter’s Webinars Program Manager, enabling the chapter to continue engaging and serving members through online events and overseeing the monthly Pivotal Webinar program.

Understand objectives

- Understand techniques for individual and group learning
- Understand techniques for machine learning
- Apply various learning techniques to accelerate your project team’s growth

Why it matters

- The pace of technological change is accelerating
- We find ourselves constantly needing to upskill
- Learning is a fundamental to who are are

Forbes: Human Skills Will Be Most Important In The Digital Future Of Work

PWC: Upskilling Hopes and Fears for 2021

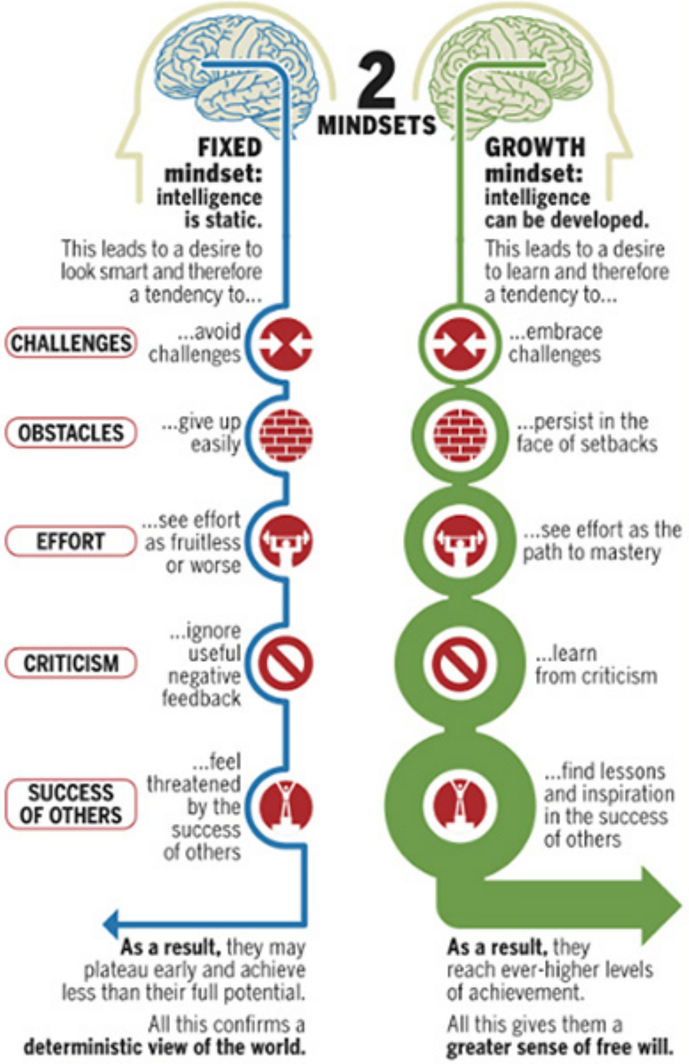
Resources for learning

- Classes (formal instruction)
- Books / Articles
- Videos (YouTube)
- Podcasts
- Guided learning apps (LinkedIn Learning, Udemy, Code Academy, Pluralsight, Khan Academy, Degreed)

Where do you want to grow?

- What’s your passion (your WHY)?
- What are your goals?
- What skills are going to help you achieve your goals?
- Mix of technical and behavioral skills
- How much to learn? What level of skill do you need?
- What do you hope to do with the new skills?

Growth vs Fixed Mindset



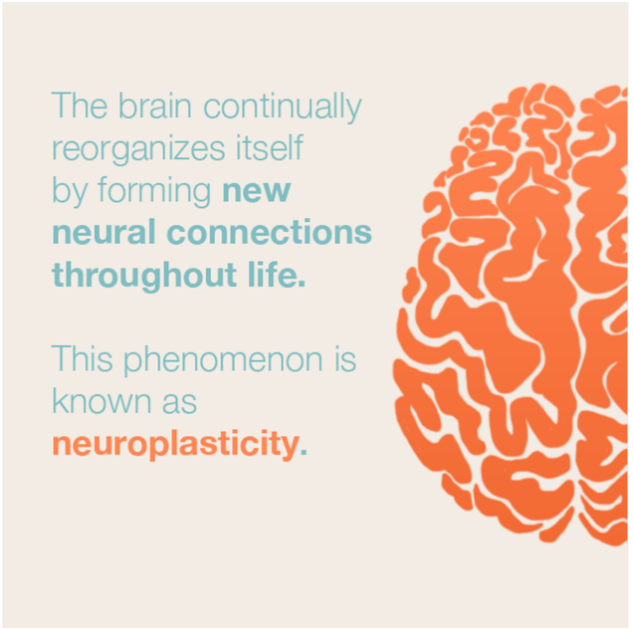
Our prevailing system of management has destroyed our people. People are born with intrinsic motivation, self-respect, dignity, curiosity to learn, joy in learning. The forces of destruction begin with toddlers—a prize for the best Halloween costume, grades in school, gold stars—and on up through the university. On the job, people, teams, and divisions are ranked, reward for the top, punishment for the bottom. Management by Objectives, quotas, incentive pay, business plans, put together separately, division by division, cause further loss, unknown and unknowable.

-W. Edwards Deming

Source: "Mindset", Dr. Carol Dweck

How humans learn

- Neuroplasticity - the ability for neural networks to grow and reorganize that results in learned patterns

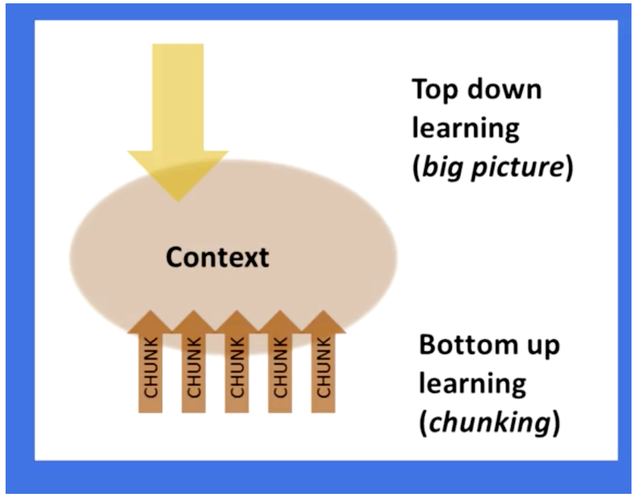




Source: "Neuroplasticity", Stephanie Liou, Stanford University

How humans learn

- Chunking
  - Creating bits of learned neural connections
  - These can be combined into progressively larger, more complex chunks



Sources: “A Mind for Numbers”, Barbara Oakley, Ph.D.; “Learning How to Learn”: Coursera course, Oakley & Dr. Terrence Sejnowski

Techniques for learning

- Retrieval - actively try to recall information
  - Look away and test yourself
  - Electronic flashcards (Anki) - spaced repetition
  - Feynman technique - Write out a topic on a piece of paper, then explain it as simply as you can. Find any weak points, and go back and strengthen your ability.
- Feedback - timely and relevant information on how we're doing
  - Outcome feedback - Are you doing better or not?
  - Informational feedback - What are you doing wrong?
  - Corrective feedback - How can you fix what you're doing wrong?

Sources: "A Mind for Numbers", Barbara Oakley, Ph.D.; "Ultralearning", Scott H. Young; "How We Learn", Benedict Carey

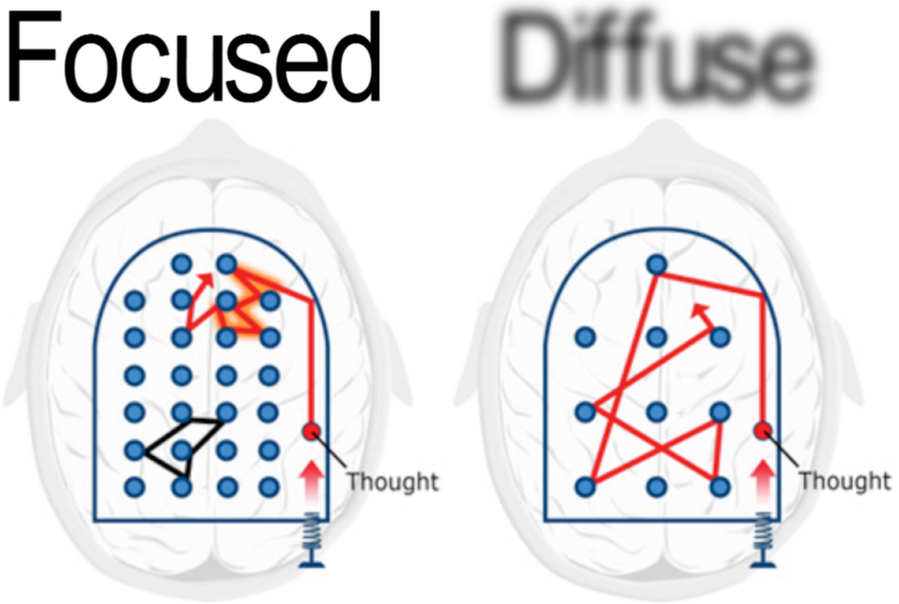
Techniques for learning

- Drilling - Practice on weaker areas
- Interleaving - Practice in different contexts
- Experimentation - Trying different approaches

Sources: “A Mind for Numbers”, Barbara Oakley, Ph.D.; “Ultralearning”, Scott H. Young; “How We Learn”, Benedict Carey

Techniques for learning

- Focus - Learning requires focused attention
- Intuition - Relaxed, ‘diffuse’ mindset helps us generate new ideas and make new connections
- It is the interplay between these two where we learn most effectively



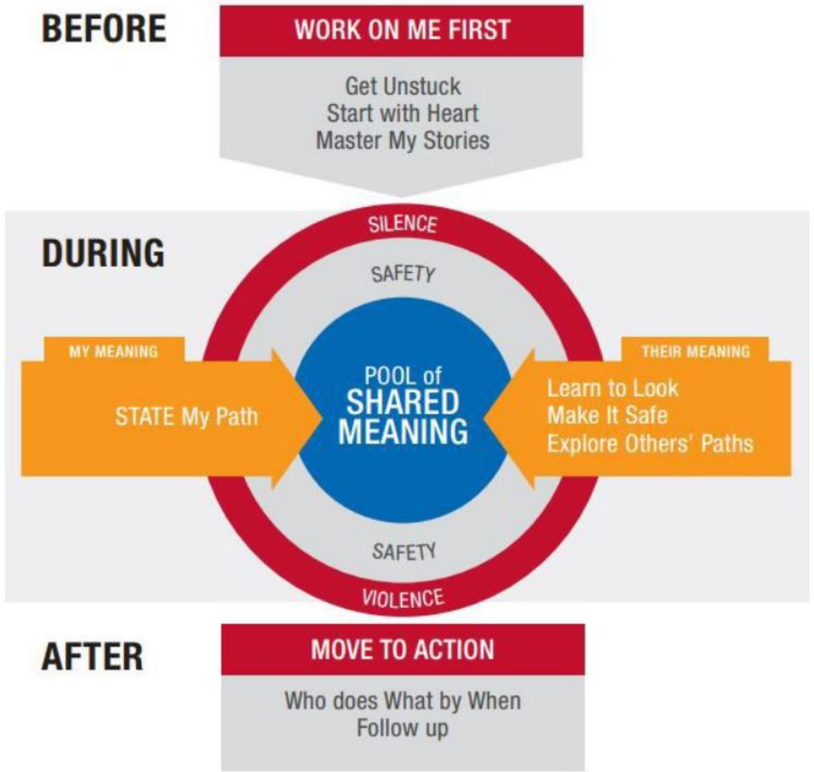
Sources: “A Mind for Numbers”, Barbara Oakley, Ph.D.; “Ultralearning”, Scott H. Young

Behavioral skills

- Examples of behavioral skills applicable to projects: Emotional intelligence, task management, negotiation, delegating, mindfulness, etc.
- Use existing frameworks by experts in the field
- Build mental models for how to approach the situation
- Practice is key

Behavioral skills

- Example: Crucial conversations



Source: Image from Boston College; Crucial Conversations by Kerry Patterson et. al

Pitfalls and best practices

- Pitfalls
  - Lack of sleep
  - High stress
  - Lack of focus
  - Illusion of competence
- Best practices
  - Well rested
  - Moderate stress
  - Soft, focused awareness
  - Emphasize *understanding*

What about visual vs auditory learners?

- There is no scientific evidence that we are ‘visual’, ‘auditory’, or ‘tactile’ learners
- We are not binary in our preference for learning
- Rather, we frequently operate in all three
- Believing that we are one type of learner vs another can harm our ability to learn through other means

Sources: “There’s no such thing as ‘auditory’ or ‘visual’ learners”, Business Insider; “Opinion: Are you a visual or auditory learner? It doesn’t matter.” New York Times; “Neuromyths: No, you aren’t a ‘visual’ or ‘auditory’ person”, Inverse

Exercise: Individual learning

- Breakout rooms of 4-5 people for 10 minutes
- Discuss the following:
  - What approaches do you and your teams use to learn new skills relevant to your projects?
  - Based on the techniques discussed, what change would you make to learn more effectively?